

OKHA, Russian Federation

WMO: 320100

Lat: 53.518N

Lon: 142.880E

Elev: 121

StdP: 14.63

Time Zone: 11.00 (E11)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-20.4	-16.8	-25.0	1.4	-20.1	-21.4	1.7	-16.1	32.6	8.5	29.6	7.5	4.3	310	0.773

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.3	77.5	64.3	73.9	62.4	70.2	61.0	66.1	73.9	64.0	71.2	62.1	68.8	10.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.1	86.8	69.8	61.1	80.9	67.2	59.1	75.2	65.2	31.0	73.5	29.4	71.1	28.0	68.7	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.9	23.3	20.4	DB	-29.6	85.8	4.3	3.6	-32.7	88.4	-35.2	90.5	-37.7	92.5	-40.8	95.1	(4)
(5)			WB	-29.1	69.4	4.1	1.4	-32.0	70.4	-34.5	71.3	-36.8	72.1	-39.8	73.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.7	0.9	0.9	11.1	27.5	38.5	50.5	57.7	59.3	51.2	37.3	21.6	9.3		
	DBStd	22.33	11.79	10.28	8.89	6.93	6.46	8.37	6.01	5.13	6.41	6.03	9.32	10.85		
	HDD50	7814	1523	1374	1205	676	364	95	8	1	61	394	852	1261		
	HDD65	12565	1988	1794	1670	1126	821	441	236	189	414	858	1302	1726		
	CDD50	755	0	0	0	0	8	110	248	290	98	1	0	0		
	CDD65	31	0	0	0	0	0	6	11	13	1	0	0	0		
	CDH74	302	0	0	0	0	0	85	105	108	4	0	0	0		
	CDH80	37	0	0	0	0	0	13	14	10	0	0	0	0		
	WSAvg	9.8	11.5	9.6	9.0	10.3	9.4	9.1	9.0	8.4	8.4	10.0	10.9	12.0		
	PrecAvg	22.6	1.4	1.0	1.2	1.1	1.5	1.4	2.0	2.6	2.3	2.7	2.9	2.4		
(15) Precipitation	PrecMax	30.9	3.8	3.0	3.9	2.4	4.6	4.0	5.0	5.0	5.7	4.4	4.0	5.3		
	PrecMin	14.1	0.1	0.1	0.2	0.4	0.1	0.2	0.1	0.6	0.5	0.7	1.1	0.6		
	PrecStd	4.6	0.8	0.7	0.9	0.5	1.0	0.9	1.4	1.3	1.3	0.9	0.9	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.3	29.1	35.3	52.3	68.8	81.7	82.3	81.0	73.4	58.3	42.2	32.9		
		MCWB	29.6	28.2	30.3	42.7	51.8	62.9	66.1	66.1	62.4	49.1	40.9	32.1		
	2%	DB	28.6	27.0	30.6	45.5	61.2	75.9	77.0	77.0	68.6	52.6	39.4	30.9		
		MCWB	27.9	26.3	28.5	38.4	48.9	61.5	64.9	65.0	60.0	46.9	38.3	30.6		
	5%	DB	26.6	23.0	28.7	41.1	55.9	70.7	73.1	73.2	64.7	49.3	36.6	28.5		
		MCWB	26.2	22.3	27.1	35.4	45.7	58.5	61.8	63.4	57.8	45.3	35.3	27.8		
	10%	DB	21.9	17.8	26.1	36.7	51.1	65.3	68.9	69.6	61.0	46.6	33.9	26.1		
		MCWB	21.3	17.0	24.6	32.8	43.7	55.6	60.9	62.2	55.8	43.7	32.7	25.4		
	0.4%	WB	29.9	28.9	30.9	43.0	53.9	65.2	68.2	69.1	64.6	51.1	41.9	32.5		
		MCDB	30.1	29.1	33.3	49.6	66.3	77.7	79.0	77.3	69.7	54.4	42.3	32.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.1	26.4	29.2	39.2	49.5	62.2	65.7	66.6	61.5	48.2	38.5	30.7		
		MCDB	28.6	27.0	30.6	45.6	59.1	74.9	73.9	73.6	66.8	51.3	39.2	30.9		
	5%	WB	26.1	22.5	27.2	35.8	46.6	59.1	63.6	64.6	58.7	46.0	35.6	27.9		
		MCDB	26.6	22.9	28.6	40.2	54.2	69.7	70.4	71.4	63.4	48.4	36.3	28.2		
	10%	WB	21.1	17.0	24.6	33.2	44.1	56.5	61.7	62.6	56.6	44.0	33.0	25.4		
		MCDB	21.9	17.8	26.3	36.2	51.0	64.2	68.0	68.5	60.0	46.4	33.7	26.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	15.1	17.6	12.0	13.7	17.3	14.5	14.3	13.8	9.8	10.0	10.8		
		MCDBR	10.7	13.2	17.1	20.1	26.6	28.1	24.5	24.4	20.3	17.0	9.3	12.1		
		MCWBR	10.6	13.1	14.5	13.3	15.0	13.9	12.8	12.6	11.2	11.6	8.3	11.1		
		MCDBR	10.5	12.8	13.8	17.9	24.1	26.1	20.7	20.1	17.1	14.0	10.0	11.2		
	5% WB	MCWBR	10.5	12.7	12.1	12.0	14.3	13.8	11.7	11.6	10.3	10.5	9.5	11.4		
(40) Clear-Sky Solar Irradiance	taub		0.286	0.301	0.362	0.461	0.502	0.466	0.474	0.424	0.377	0.379	0.343	0.298		
	taud		2.007	1.998	1.862	1.690	1.818	2.105	2.140	2.304	2.377	2.168	2.074	1.998		
	Ebn at Noon		202	244	250	238	238	250	246	252	251	213	177	168		
	Edh at Noon		24	34	49	67	62	47	45	36	30	29	23	21		
(44) All-Sky Solar Radiation	RadAvg		303	652	1092	1469	1565	1712	1484	1256	990	535	257	181		
	RadStd		43	65	99	158	157	201	179	114	79	47	21	41		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	-248	+110	N/A

Nomenclature: See separate page